

Comunicación SPI MAX6675

Ciclos mínimos

Tiempo clk_SPI $\rightarrow$ $\lim_{\text{inf}} 4,3 \text{ MHz}$ $\lim_{\text{sup}} 3 \text{ MHz}$

$t_{\text{sp}} = 700 \text{ ns}$ $\rightarrow$ min

$$N \text{ ciclos} = \frac{25}{4} = 6.25$$

$$\frac{25}{7} = 3.57 \text{ MHz}$$

$$\frac{25}{8} = 3.125 \text{ MHz}$$

$$t_{\text{CH}} = t_{\text{CL}} = \frac{7}{2(3.125 - 10^6)} = 16 \cdot 10^{-8} = 160 \text{ ns}$$

$$t_{\text{sp}} = 700 \text{ ns}$$

con 4 ciclos:

$$400 \text{ ns}(4) = 1600 \text{ ns} > 700 \text{ ns}$$

ELECTRICAL CHARACTERISTICS (continued)

(V_{CC} = +3.0V to +5.5V, T_A = -20°C to +85°C, unless otherwise noted. Typical values specified at +25°C.) (Note 1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Output High Voltage	V _{OH}	I _{SOURCE} = 1.6mA	V _{CC} - 0.4			V
Output Low Voltage	V _{OL}	I _{SINK} = 1.6mA		0.4		V
TIMING						
Serial Clock Frequency	f _{SCL}			4.3		MHz
SCK Pulse High Width	t _{CH}		100			ns
SCK Pulse Low Width	t _{CL}		100			ns
CSB Fall to SCK Rise	t _{CSB}	C _L = 10pF	100			ns
CSB Fall to Output Enable	t _{COV}	C _L = 10pF		100		ns
CSB Rise to Output Disable	t _{TR}	C _L = 10pF		100		ns
SCK Fall to Output Data Valid	t _{DO}	C _L = 10pF		100		ns

Note 1: All specifications are 100% tested at T_A = +25°C. Specification limits over temperature (T_A = T_{MIN} to T_{MAX}) are guaranteed by design and characterization, not production tested.

Note 2: Guaranteed by design. Not production tested.

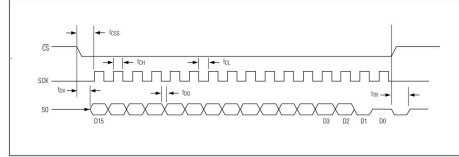
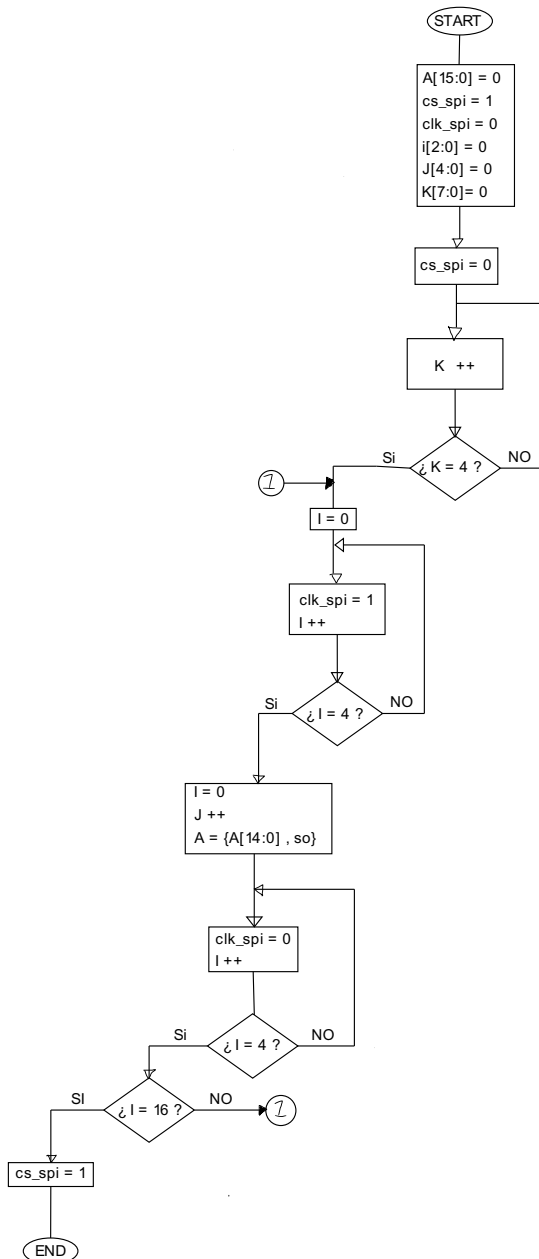


Figure 1b. Serial Interface Timing

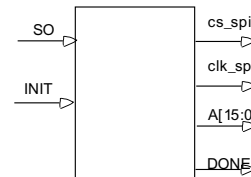
BIT	DUMMY SIGN BIT	12-BIT TEMPERATURE READING												THERMOCOUPLE INPUT	DEVICE ID	STATUS
		Bit	15	14	13	12	11	10	9	8	7	6	5			
	0	MSB												LSB	0	Three state

Figure 2. SO Output

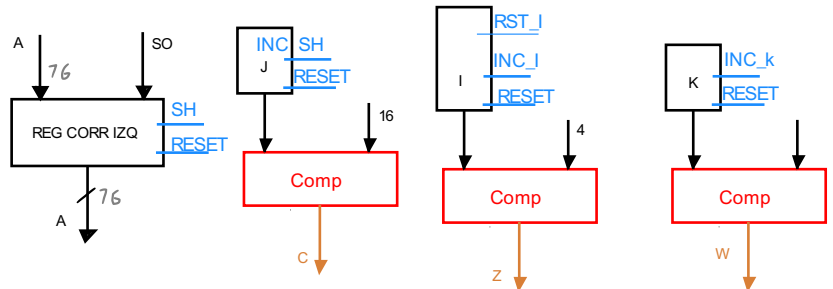
Diagrama de flujo



Caja Negra



Camino de datos



Bloque de control

